

JEE MAIN REVISION MASTER: STATES OF MATTER

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Real Gases & van der Waals Equation	7 PYQs	Real Gases
Standard	Ideal Gas Laws, Dalton's Law & Graham's Diffusion	1 PYQs	Gas Laws

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Real Gases & van der Waals Equation	$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT, \quad Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}, \quad T_c = \frac{8a}{27Rb}$
Ideal Gas Laws, Dalton's Law & Graham's Diffusion	$PV = nRT, \quad P = \frac{dRT}{M}, \quad \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \frac{V_1 / t_1}{V_2 / t_2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Real Gases & van der Waals Equation (Moderate Yield — 7 PYQs)

Core Principle: Real gas equation: $\left(P + \frac{a}{V_m^2}\right)(V_m - b) = nRT$. Constant a measures magnitude of intermolecular attractive forces; constant b measures effective molecular volume (co-volume $b = 4 V_m$). Compressibility factor $Z = (PV / nRT)$: for ideal gas $Z=1$; at low pressure $Z < 1$ (attraction dominates); at high pressure $Z > 1$ (repulsion dominates). Real gases approach ideal behavior at HIGH temperature and LOW pressure.

■ **Caution / JEE Pitfall:** For Hydrogen and Helium, attractive forces are negligible ($a \approx 0$), so $Z = 1 + (Pb / RT) > 1$ at all ordinary temperatures!

■ **Speed Trick:** A gas with higher critical temperature T_c and higher a value is more easily liquefied (e.g. NH_3 , SO_2 liquefy more easily than N_2 , H_2).

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the wrong relation for real gases :..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The volume of 2.8g of CO at 27°C and 0.821 atm pressure is ($R=0.08210\text{L. atm Kmol}^{-1}$)..." [Ans: N/A]

3.2 Ideal Gas Laws, Dalton's Law & Graham's Diffusion (Standard Yield — 1 PYQs)

Core Principle: Ideal gas equation: $PV = nRT = (w/M)RT \implies PM = dRT$. Dalton's law: $P_{\text{(total)}} = \sum P_i$, where $P_A = \chi_A P_{\text{(total)}}$. Graham's law: rate of diffusion $r \propto \frac{1}{\sqrt{M}} \propto \frac{1}{\sqrt{d}}$.

■ **Caution / JEE Pitfall:** Graham's law uses the inverse square root of molar mass: lighter gas diffuses faster (e.g. H_2 diffuses 4 times faster than CO_2 because $\sqrt{32/2} = 4$).

■ **Speed Trick:** Gas density formula $d = (PM / RT)$. At higher temperature and lower pressure, gas density is minimum.

Representative Source PYQs:

- **[2019 Shift] Q#1 (MCQ):** "Which of the following equations does not represent Charles's law for a given mass of gas at constant pressure?..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2022 (Q#2)

Which property of CO_2 makes it biologically and geo-chemically important?

- (A) Its colourless and odourless nature
- (B) Its low solubility in water
- (C) Its high compressibility
- (D) Its acidic nature

Correct Answer: None

Solution Analysis: The property of low solubility of carbon dioxide in water makes it biologically and geo-chemically important. Carbon dioxide on reaction with water form carbonic acid, which dissociates to give HCO_3^- ions. H_2CO_3/HCO_3^- buffer system helps to maintain pH of blood between 7.26-7.42.

Benchmark Problem 2: KCET 2019 (Q#1)

Which of the following equations does not represent Charles's law for a given mass of gas at constant pressure?

- (A) $(V / T) = K$
- (B) $\log V = \log K + \log T$
- (C) $\log K = \log V + \log T$
- (D) $(d(\ln V) / dT) = (1 / T)$

Correct Answer: None

Solution Analysis: Charles's law states that "pressure remaining constant, the volume of a fixed mass of a gas is directly proportional to its absolute temperature. The general equation is as follows : $(V / T) = \text{Constant}(K); V = KT$ (i) Taking the logarithm of equation (i) $\log V = \log K + \log T$ (ii) $\log K = \log V - \log T$ (iii) Equation (ii) can also be represented as $\ln V = \ln K + \ln T$ (iv) Taking differentiation of equation (iv) w.r.t T . $(d(\ln V) / dT) = 0 + (1 / T); (d(\ln V) / dT) = (1 / T)$